



AI-Powered Multimodal Churn Prediction & Sentiment Analysis

Enterprise Control Panel (EvoBoost Classification + Advanced NLP Aspect Mappings) [↔](#)

Executive Analytics Matrix Agentic Remediation Center



Deep Data Exploration & Predictive Signals

Macro-level overview of correlation variables, feature weights, and user sentiment drivers.

Total Monitored Directory Space

400 Accounts

Calculated Model Churn Rate

28.0%

↓ -2.4% vs Last Quarter

Monthly Revenue At Risk

\$9,507.97

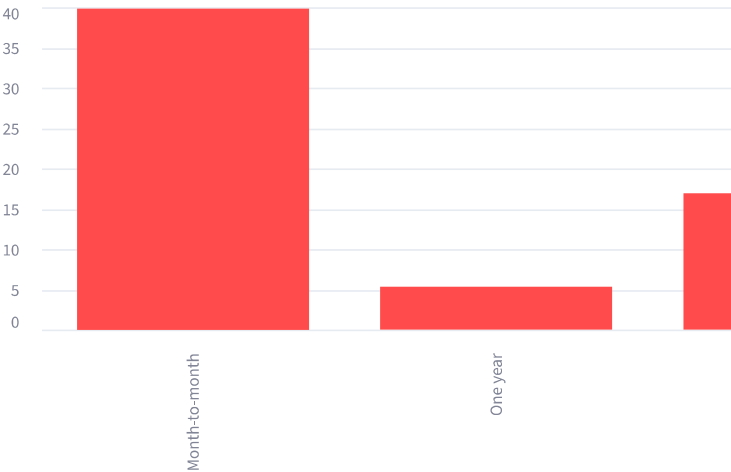
↑ Critical Target Set

Global Frustration Baseline

3.9%



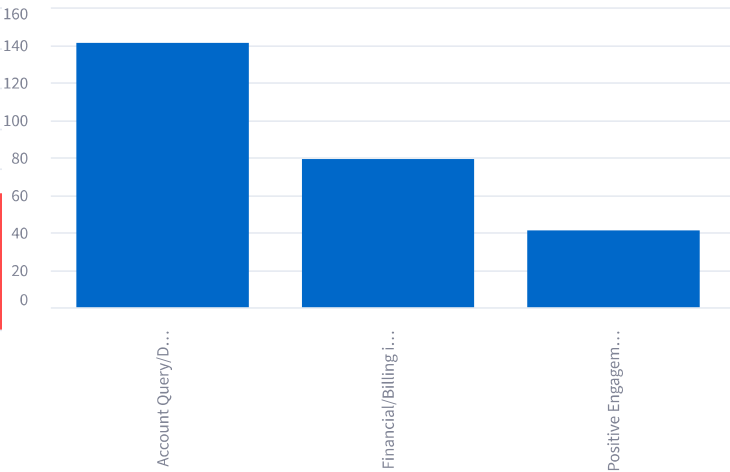
Churn Rate Dynamics by Contract Class



Insight: Short term Month-to-month contracts represent the highest density churn threat space.



Support Aspect Categorization Volume



Insight: Structural categorization derived completely offline using zero-shot semantic matching maps.



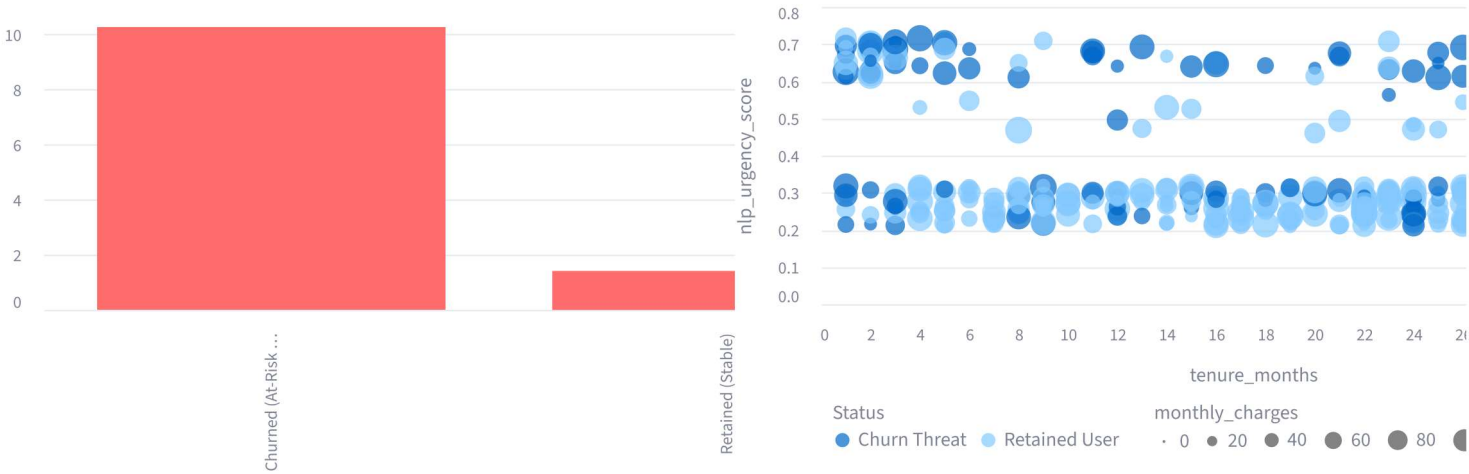
Sentiment Intensity Profiles: At-Risk vs Retained Cohorts



Mean Customer Frustration Levels by User Status



Continuous Escalation Urgency Score Matrix Profile



Plot Interpretation: Higher cluster distribution coordinates represent critical system intervention zones (Short tenure, high financial footprints, and severe linguistic urgency metrics).

Custom EvoBoost Model Validation Performance

Exact machine learning performance benchmarks computed over the out-of-sample holdout validation split ($N_{\text{test}} = 100$).

Model Accuracy ?	Model Precision ?	Model Recall (Sensitivity) ?	Model F1-Score ?
84.00%	77.27%	60.71%	68.00%

Validation Confusion Matrix Topology

	Predicted Retained (0)	Predicted Churned	Show Analytical Formulations
Actual Retained (0)	67	5	Accuracy = $\frac{TP + TN}{\text{Total}} = \frac{17 + 67}{100} = 84.00\%$ Precision = $\frac{TP}{TP + FP} = \frac{17}{17 + 5} = 77.27\%$ Recall = $\frac{TP}{TP + FN} = \frac{17}{17 + 11} = 60.71\%$
Actual Churned (1)	11	17	

Interpretation Note: A low false alarm footprint (5 False Positives) preserves retention budget resources.